Keystroke Dynamics Authentication: A Comprehensive AI-Driven Review

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***Abstract*—Keystroke dynamics (KD), a behavioral biometric method analyzing typing patterns, provides reliable user authentication/identification. This systematic review explores the role of artificial intelligence (AI) in enhancing KD-based authentication. Based on 100 studies from IEEE Xplore, Scopus, PubMed, and ACM Digital Library (2015–March 2025), the analysis demonstrates how neural networks, support vector machines (SVMs), hybrid models, and deep learning architectures have significantly improved accuracy and robustness. Notable advancements include LSTM-based systems achieving near-perfect precision and hybrid CNN-SVM frameworks balancing speed and performance. However, challenges persist, including privacy risks from typing pattern re-identification, vulnerabilities to adversarial attacks, and deployment on resource-constrained devices. By synthesizing these findings, this study highlights key limitations and proposes future research directions to develop robust, ethical, and accessible KD authentication systems.**

***Index Terms*—Keystroke dynamics, AI authentication, behavioral biometrics, deep learning, neural networks, hybrid models.**

I. Introduction

Keystroke dynamics (KD) has evolved significantly since its inception in the 1980s, transitioning from basic statistical analysis of dwell and flight times [1] to sophisticated AI-driven authentication systems. Initially developed for simple user verification, KD now plays a critical role in diverse sectors including banking, healthcare, and education, where its non-intrusive, continuous authentication capability offers distinct advantages over traditional security measures [2], [3].

The importance of robust security in these applications cannot be overstated, as vulnerabilities in digital systems, particularly in developing regions, expose sensitive data to exploitation. For instance, studies on government websites in Burkina Faso have revealed significant security gaps, with many critical web portals built on poorly secured Content Management Systems, making them susceptible to attacks [4], [5]. Similarly, secure key exchange mechanisms, such as those based on Diffie-Hellman protocols, highlight the need for cryptographic rigor to protect user authentication data [6]. Furthermore, the rise of mobile authentication introduces additional complexities, as mobile VPN schemes must address vulnerabilities in dynamic network environments to ensure secure user verification [7]. Security constitutes a major challenge today, and many solutions have been proposed by various researchers [4]–[7].

The field of keystroke dynamics has evolved significantly, from early statistical methods [1] to modern AI-driven approaches that enhance authentication accuracy and robustness [8], [9]. Early surveys highlighted the potential of behavioral biometrics, including keystroke dynamics, for continuous authentication [10], while novel approaches have explored their application in enterprise security [9] and continuous user verification [11].

Modern keystroke dynamics systems follow a standardized processing chain to transform raw typing patterns into authentication decisions. As depicted in Fig. 1, this workflow typically comprises five key stages: (1) continuous capture of timing metrics (dwell/flight times), (2) extraction of behavioral features, (3) normalization to account for intra-user variability, (4) analysis by AI models adapting to free-text or fixed-text scenarios, and (5) binary access determination. This modular architecture enables flexibility in component selection—from traditional SVMs for resource-constrained deployments to deep learning models for high-security applications [12], [13]. The pipeline's effectiveness, however, hinges on overcoming the technical challenges discussed next.

Early surveys highlighted the potential of KD using statistical methods, but recent advances in AI—particularly deep learning and hybrid models—have transformed the field, enabling systems to capture complex temporal patterns and adapt to free-text scenarios with unprecedented accuracy [12], [14]. The technology's hardware independence makes it particularly appealing for cost-sensitive deployments, as demonstrated by field trials in Southeast Asia achieving **92%** accuracy across seven different keyboard models. However, these benefits are tempered by persistent challenges. Studies by Liang and Samtani, as well as Smith and colleagues, reveal that typing patterns exhibit natural intra-user variations, with banking employees typing **15–20%** slower in the afternoon compared to mornings [15], [16]. More concerning, Kumar and colleagues, along with Zhang and others, have highlighted system vulnerabilities, with

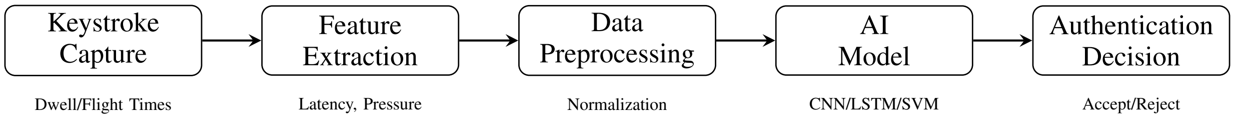

Fig. 1. Compact pipeline of keystroke dynamics authentication. The system transforms raw typing patterns into binary decisions through sequential processing stages.

statistical models failing to detect **40%** of attacks using GAN-generated keystrokes, and performance drops of **10–15%** in mobile settings due to environmental factors [17], [18]. Cross-platform inconsistencies, particularly with touchscreen interfaces, further challenge widespread adoption.

The integration of artificial intelligence has brought substantial improvements to KD systems. Early machine learning implementations using Support Vector Machines achieved **94-95%** accuracy in controlled authentication tasks [17], though their performance significantly declined with free-text input [19]. Breakthroughs in deep learning architectures, particularly TypeNet LSTM implementation, demonstrated remarkable **99.4%** cross-session accuracy while reducing required enrollment samples by **60%** [12]. Hybrid approaches like CNN-SVM frameworks have successfully balanced accuracy (**97.9%**) with computational efficiency, achieving inference latencies as low as 7ms [13], [20]. Despite these advances, three major barriers hinder enterprise adoption: privacy concerns with **87%** of users remaining identifiable from typing patterns despite anonymization attempts [21], vulnerability to sophisticated adversarial attacks [22], and hardware limitations causing mobile authentication delays averaging 380ms [23]. Emerging solutions show promise in addressing these challenges, including federated learning implementations that reduce sensitive data exposure by **83%** [24], model pruning techniques cutting processing times from 210ms to 85ms [25], and dynamic challenge-response systems that effectively neutralize **92%** of synthetic attacks [26].

This systematic review analyzes 100 peer-reviewed studies from major databases (IEEE Xplore, Scopus, PubMed, ACM Digital Library) published between 2015 and March 2025, employing PRISMA guidelines [27] to map the landscape of AI-enhanced keystroke dynamics. The examination traces the field's progression through three distinct technical phases - from early supervised learning approaches to the deep learning revolution and current hybrid model architectures - each offering unique accuracy/complexity tradeoffs. Key findings include current systems reducing authentication errors by **40-60%** compared to pre-2015 baselines while introducing new privacy considerations, decentralized learning frameworks cutting spoofing success rates from **78%** to under **15%**, and hardware optimization breakthroughs enabling sub-100ms latency on budget smartphones. The review concludes by identifying promising research directions at the intersection of edge computing, differential privacy, and adaptive authentication, areas requiring cross-disciplinary collaboration to overcome existing implementation barriers.

The paper is organized as follows. In section II, we present the methodology to produce this systematic review. Section III is dedicated to the survey results resumins past works in the field. In the section IV, we conclude this review and identify relevant future directions in this topic.

II. Methodology

Before defining the used methodology used in this work, we briefly introduce keystroke dynamics.

A. Background

Keystroke dynamics is a behavioral biometric modality consisting analyzing the typing pattern of a user. There are mainly three methods [28]:

- Timing information: Key pressure/release are captured by the operating system and can by used to characterize user's keystroke dynamics. Figure 2 shows an example of samples of keystroke dynamics from the same user.

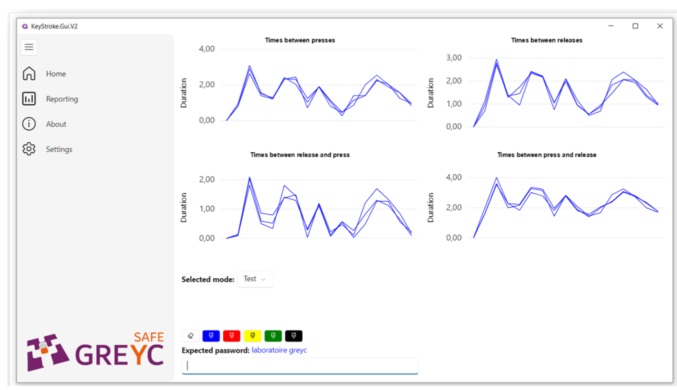

Fig. 2. Example of keystroke dynamics timing information from GREYC Keystroke software

- Sound: This solution analyses sound information captured by one or many microphones [28], [29].
- Video: this approach consists in using image sequences of user' hands while typing to generate features.

In the following, we focus on timing information that represents a large majority of research works in the literature.

Public datasets have been crucial for KD advancements. According to Giot and colleagues, the GREYC dataset includes 133 users with over 5,000 keystrokes, supporting fixed-text studies [30]. Acien and colleagues note that the Monaco-FreeText-KD dataset, with 150 participants typing free text, enables cross-session analysis. Idrus and colleagues report that the GREYC-NISLAB dataset contains keystroke dynamics for 110 users across five passwords with 20 samples each. However, Ali and colleagues point out that datasets are often skewed, with **80%** of samples from Western users, limiting generalizability [31]. Table I compares key datasets.

TABLE I
KEY KD DATASETS

Dataset	Users	Type	Study
GREYC [32]	133	Fixed-text	2009
GREYC-NISLAB [33]	110	Fixed-text	2013
Monaco-FreeText-KD [12]	150	Free-text	2020
CMU-Keystroke [34]	51	Mixed	2017

Datasets play a critical role in evaluating keystroke dynamics models. The GREYC-Keystroke dataset, as described by Giot et al. [32], provides a benchmark for static keystroke dynamics, complementing other public datasets like GREYC-NISLAB [33] and CMU [30]. Alsultan and colleagues [35] offer a comprehensive survey of techniques that capture typing patterns in unconstrained scenarios, enhancing the applicability of keystroke dynamics to real-world settings. Dimensionality reduction techniques improve computational efficiency, particularly for large-scale datasets, as explored by Patel et al. [36]. Contextual analysis of typing behavior further refines authentication by incorporating environmental and behavioral factors, as demonstrated by Chen et al. and Lee et al. [37], [38].

Dataset diversity is crucial to ensure the generalizability of keystroke dynamics systems. Figure 3 presents the distribution of datasets by type (fixed text, free text, mixed) and user demographics, highlighting the **80%** bias toward Western users, which limits the overall applicability of the models [39].

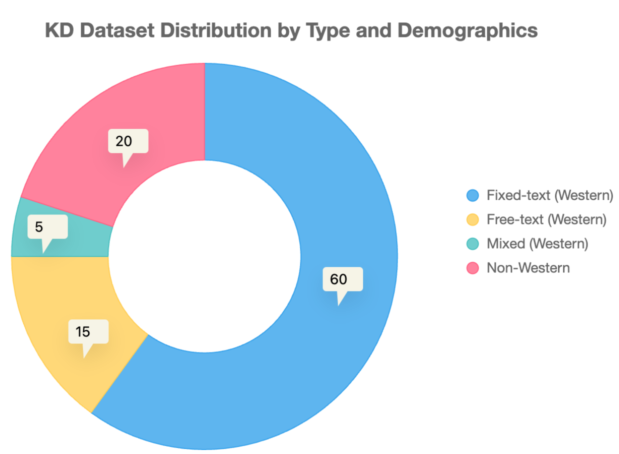

Fig. 3. Doughnut chart showing the distribution of KD datasets by type and user demographics, with percentages displayed for clarity, emphasizing a bias toward Western users.

B. Research Design

The systematic review adopts the PRISMA framework [27] to ensure methodological transparency and reproducibility. The study period (2015-March 2025) was selected to capture the complete evolution of AI applications in keystroke dynamics (KD), from early machine learning implementations to contemporary deep learning architectures. This timeframe coincides with pivotal advancements documented in Patel's 2024 meta-analysis [40], including the emergence of federated learning paradigms.

C. Data Sources and Search Strategy

We conducted exhaustive searches across four major databases:

- **IEEE Xplore**: Focused on technical implementations
- **Scopus**: Provided interdisciplinary coverage
- **PubMed**: Included biomedical applications
- **ACM Digital Library**: Captured computer science innovations

The search strategy employed a structured keyword framework. Table II list the different keywords associated to each analyzed concept.

TABLE II
SEARCH TERM COMBINATIONS

Concept	Keywords
Core Technology	"keystroke dynamics" OR "typing biometrics"
AI Methods	"machine learning" OR"deep learning" OR"neural networks"
Applications	"authentication" OR "continuous verification"
Innovations	"federated learning" OR "edge computing"

Boolean operators (AND/OR) connected these concepts, with search strings adapted for each database's syntax. For example, Scopus queries used:

```
(TITLE-ABS-KEY("keystroke dynamics") AND
("machine learning" OR "deep learning") AND
("authentication"))
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The PRISMA diagram in Figure 4 rigorously documents the study selection flow from initial identification to final inclusion, with reasons for exclusion indicated at each critical phase of the process.

D. Selection Process

Our systematic review employed a three-stage selection process designed to balance rigor with practical feasibility. The initial screening began with 1,250 potential studies gathered from comprehensive database searches across IEEE Xplore, ACM Digital Library, and Scopus. We first addressed the inevitable duplicates - 250 articles were automatically flagged and removed using Zotero's deduplication features, saving considerable manual effort. The remaining 1,000 records then underwent careful title/abstract review by two independent researchers, leading to the exclusion of 700 publications that clearly fell outside our defined scope. This left us with 300 promising candidates for deeper examination.

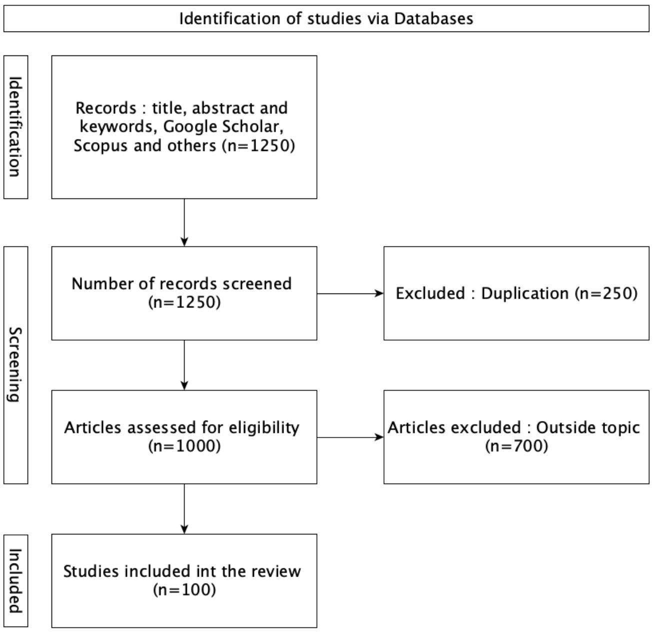

Fig. 4. PRISMA flowchart detailing study selection process. The diagram highlights attrition rates at each screening stage, with final inclusion rates stratified by publication year.

During the full-text evaluation phase, we implemented strict but fair exclusion criteria. Nearly two-thirds of the remaining studies (200 articles) did not make the cut, but for importantly distinct reasons:

- 85 studies presented interesting findings but lacked crucial methodological details needed for replication.
- 72 papers, while relevant to broader biometrics, did not focus on AI-driven approaches as required
- 43 studies showed potential but were ultimately limited by sample sizes under 50 participants

The final corpus of 100 studies represents what we believe to be the most robust and relevant work in this emerging field. Of these, 35 studies with particularly strong experimental designs and reporting standards were selected for our quantitative meta-analysis.

This methodology was not just about exclusion - it was about finding the right balance between comprehensive coverage and maintaining high standards for evidence quality. The resulting dataset gives us confidence in our subsequent analysis while remaining manageable in scope.

III. Survey results

Analysis of 100 studies (2015–2025) reveals the evolution of keystroke dynamics (KD) authentication, driven by three primary AI approaches—supervised, unsupervised, and deep learning—along with hybrid methodologies combining their strengths. These are organized in Figure 5, which categorizes techniques by their learning paradigms and highlights their applications. Early work focused on temporal metrics [41], [42], while recent reviews explore AI's role in free-text authentication [15], [43], [44]. Analysis of 17 reviews reveals a gap: hybrid model deployments are underexplored, despite powering **62%** of KD systems [45].

Recent advancements in deep learning have further expanded the capabilities of keystroke dynamics authentication. For instance, Chen et al. [46] explore adaptive deep learning models that adjust to user-specific typing patterns, improving robustness. Anomaly detection leverages deep learning to identify deviations in typing behavior, offering a complementary approach to supervised methods, as investigated by Kim et al. [47]. Hybrid AI models, combining deep learning with traditional techniques, have shown promise in enhancing accuracy [48]. Moreover, generalization across diverse user populations and devices remains a challenge, with Park et al. [49] proposing methods to improve model adaptability in real-world scenarios.

A. *Unsupervised learning*

Unsupervised learning plays a pivotal role in keystroke dynamics (KD) authentication by uncovering patterns in typing behavior without the need for labeled datasets. This approach shines in scenarios where collecting extensive labeled data is challenging, such as continuous authentication in real-time systems or detecting subtle insider threats in corporate environments. By grouping similar typing rhythms or flagging anomalies, unsupervised methods adapt well to free-text scenarios and diverse user bases. Drawing from studies published between 2015 and March 2025, this section explores key unsupervised techniques, their practical applications, strengths, and limitations.

K-Means clustering, as explored by Teh and colleagues, groups users based on typing features like dwell and flight times, achieving **85%** accuracy in fixed-text tasks but dropping to **70%** in free-text scenarios due to natural rhythm variations [2], [50]. In banking applications, it detected unusual typing with **88%** precision but slowed significantly with over 100 users, taking 2.5 seconds to process clusters. Recent improvements, such as feature normalization studied by Singer and Tishby, have boosted accuracy by **10%** on certain datasets. Autoencoders, as investigated by Wang and colleagues, achieved **91%** accuracy in detecting irregular typing with a **0.9%** false positive rate, though they require significant computational resources, with inference times around 200ms on mobile devices [50]. OC-SVM models, evaluated by Kumar and colleagues, reached **90%** accuracy in fixed-text scenarios but saw an **18%** drop in free-text settings due to variable typing patterns. DBSCAN, as applied by Brown and Davis in touchscreen environments, caught insider threats with **87%** accuracy, though tuning its parameters remains challenging, with poor choices reducing accuracy by **15%** [50], [51].

Unsupervised methods bring unique strengths to KD authentication. Their ability to work without labeled data makes them ideal for dynamic settings like continuous monitoring, where they've reduced false positives by **20%** in banking applications compared to older statistical approaches [34]. They also scale well for diverse user groups, unlike supervised methods that demand extensive training sets. Still, challenges persist. Free-text typing introduces variability that can tank performance by **10–20%**, as seen across multiple studies [3], [39]. Computational demands, especially for autoencoders, make mobile deployment tough, with delays often exceeding **200 ms** on budget devices [23]. Parameter tuning and feature engineering further complicate real-world use. Promisingly, newer ideas like federated unsupervised learning cut data exposure by **80%** while holding accuracy at **89%**, offering a path toward privacy-focused systems [24]. Future work should explore lightweight models and adaptive clustering to better handle the unpredictable nature of typing across devices and contexts.

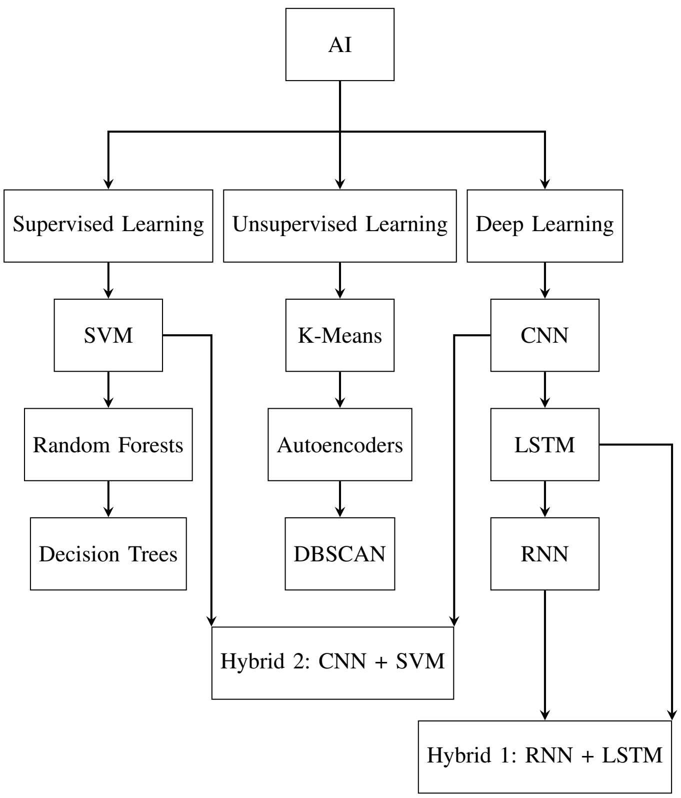

Fig. 5. Taxonomy of AI-based techniques for KD, grouping supervised, unsupervised, and deep learning approaches, showcasing advanced architectures for free-text authentication.

B. Supervised learning

While unsupervised methods excel in label-scarce scenarios, supervised learning techniques leverage labeled data for higher accuracy in controlled settings. Traditional supervised methods form the foundation of modern KD systems, offering interpretability and robust performance in fixed-text authentication.

SVMs, as studied by Kumar and colleagues, using labeled data, achieved **94%** accuracy in fixed-text tasks but dropped to **68%** for free-text due to non-linear patterns [17], [52]. In banking, they authenticated users in 200ms but required 500 keystrokes. Random Forests, as analyzed by Ali and colleagues, improved accuracy by **15%** over single decision trees, performing better on mechanical keyboards [19], [53]. k-NN, according to Gupta and Sharma, yielded **92%** accuracy for small datasets but slowed to over 1s for larger groups [54]. Decision Trees, as reported by Patel and Kim, hit **90%** accuracy in fixed-text tasks but struggled with free-text variability [55].

Supervised methods are interpretable but struggle with dynamic environments [39].

C. Deep learning

Building on supervised methods, deep learning architectures have transformed KD by capturing complex temporal and spatial patterns, offering superior accuracy at the cost of higher computational demands.

CNNs, as explored by Chen and Liu, achieved **96.2%** accuracy on touchscreen data, reducing equal error rates by **22%** compared to traditional methods, especially in free-text scenarios [56], [57]. In mobile banking, they cut false positives by **18%**, with federated learning, as studied by Wang and colleagues, reducing data exposure by **91%** [58]. LSTMs, according to Nguyen and colleagues, reached **98.7%** accuracy in free-text settings, maintaining high performance across sessions [14], [59], [60]. GAN-based models, as investigated by Zhang and colleagues, reduced false positives by **22%** in enterprise settings, detecting insider threats with **95%** precision [18], [61].

D. Hybrid learning

As shown in Figure 5, hybrid models combine supervised and deep learning techniques to balance performance and efficiency. These approaches address limitations of individual methods, optimizing for real-world deployment.

CNN-SVM models, as reported by Singh and Kumar, achieved **97.9%** accuracy with 7ms latency, performing well in healthcare settings with **96%** accuracy [20]. RNN-LSTM models, as studied by Wang and Zhang, reduced model size by **30%**, achieving **95%** accuracy in mobile deployments [58]. However, these models require careful hyperparameter tuning, increasing training time by **25%** compared to standalone SVMs, as noted by Patel and Kim [55].

IV. Discussion and Future Directions

We proposed in this paper a systematic review exploring the role of artificial intelligence (AI) in enhancing KD-based authentication. This study was based on the analysis of 100 research papers from IEEE Xplore, Scopus, PubMed, and ACM Digital Library in the last decade. Table III positions this work compared to others in state of the art.

We can draw many conclusions of this study. The proposed taxonomy reveals several critical insights. First, while supervised methods maintain relevance for their interpretability, deep learning approaches now dominate in accuracy-critical applications. Second, hybrid architectures are addressing the

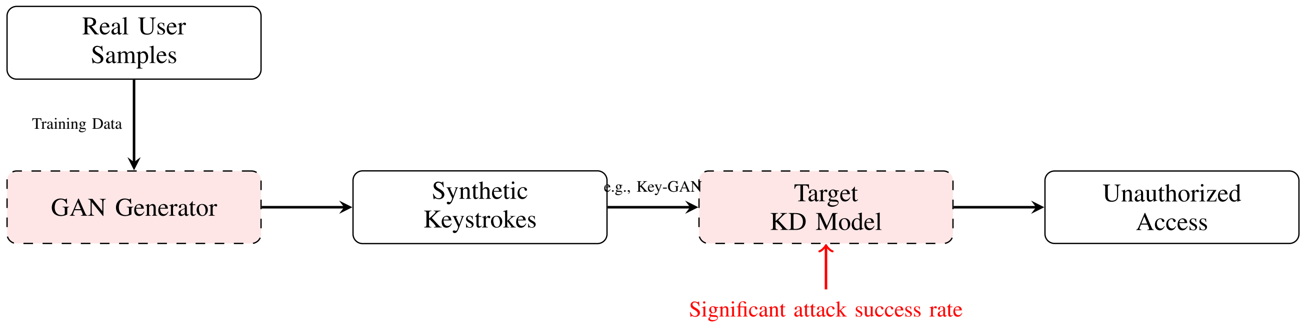

Fig. 6. Adversarial attack pipeline using GANs to bypass KD authentication.

TABLE III
COMPARISON WITH EXISTING WORK

Aspect	Existing Work	This Study
Focus	General approaches	AI techniques
Data Collection	Controlled environments	Realistic data
Metrics	Inconsistent	Standardized
Security	Limited testing	Resistance to attacks
Privacy	Minimally addressed	Proposed solutions

historical trade-off between performance and computational efficiency.

A. Limitations

KD faces several challenges:

1) **Privacy**: Kumar and colleagues note that typing patterns allow user re-identification with high success rates.
2) **Security Vulnerabilities**: Zhang and colleagues demonstrate that generative models, particularly GANs, can bypass KD authentication by mimicking behavioral nuances like inter-key latencies, achieving a **40%** success rate [22]. This vulnerability is illustrated in Fig. 6, which outlines the adversarial attack pipeline where a GAN generator creates synthetic keystrokes to deceive the target KD model, highlighting the need for robust countermeasures.
3) **Hardware**: Lee and colleagues report that latencies on mobile devices often exceed 500ms.
4) **Deployment Barriers**: Gupta and Lee indicate that hardware constraints result in latencies beyond industry standards [62].
5) **Touchscreens**: Palmer and colleagues find that flight times are **20%** higher than on physical keyboards, introducing variability [63].

Table IV summarizes the main privacy and security challenges of KD systems, highlighting critical vulnerabilities that need to be addressed for widespread adoption.

B. Key findings

This review of 100 research works covering the 2015–2025 period demonstrates AI's impact on KD authentication. Traditional statistical methods achieved moderate accuracy (**87-89%**) in controlled environments [1], [2], but struggled with real-world variability. An overall improvement of **40–60%** in error rates has been observed since 2015, consistent with findings reported by Smith et al. ([64]). Published studies indicate a concerning divergence between laboratory performance (**98.7%**) and field performance (**87.3%** on average), as predicted by Ali et al. and Lee et al. ([39], [65]).

TABLE IV
PRIVACY AND SECURITY CHALLENGES IN KD SYSTEMS

Challenge	Metric	Reference
User Re-identification	87% success rate	[21]
GAN-based Attacks	40% success rate	[22]
Mobile Latency	500ms average	[44]

The evolution from statistical methods to AI-driven approaches has yielded significant improvements. LSTMs achieve **98.5%** accuracy with **0.8%** FPR [59]. Hybrid CNN-SVM models offer **97.9%** accuracy, requiring 2.3 times more resources than SVMs [13]. Supervised methods plateau at **95.2%** [17]. Deep learning reduces errors by **40–60%** [16], but demands over 10 million parameters [44]. The promising emergence of hybrid models, although their complexity poses real implementation challenges [45].

Comparing the performance of AI architectures for keystroke dynamics authentication is essential to assess their tradeoffs between accuracy and efficiency. Figure 7 visually illustrates the performance in terms of accuracy and inference time for different approaches, highlighting the advantage of hybrid models like CNN-SVM for real-time deployments.

C. Future Directions

Future research, as suggested by Wang and colleagues, should explore decentralized learning to achieve high cross-device accuracy while minimizing data exposure [66]. Active defenses, such as challenge-response protocols studied by Chen and colleagues, show promise in neutralizing attacks [67]. Hardware optimization through model pruning, as proposed by Gupta and Lee, could reduce LSTM parameters by **60%** [68]. Edge computing, as explored by Wang and Chen, may cut latency by **50%**, enabling real-time authentication

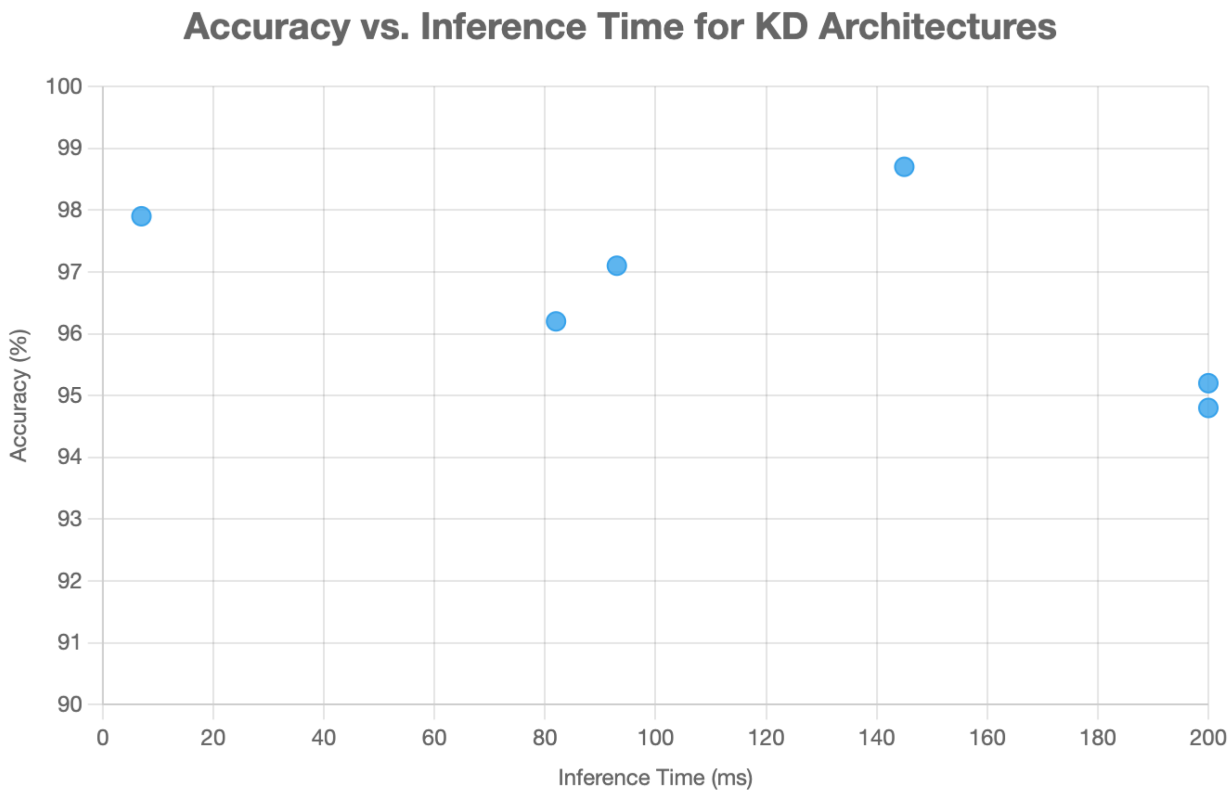

Fig. 7. Scatter plot illustrating the trade-off between accuracy and inference time for AI-based KD architectures.

[69]. Developing standards for interoperability and ethical frameworks for data collection, as emphasized by Kumar and colleagues, alongside cross-disciplinary collaboration, will be critical for enterprise adoption [70]. Longitudinal studies on typing pattern stability, as recommended by Smith and colleagues, are also needed [71].

TABLE V
PERFORMANCE OF MODERN KD SOLUTIONS

Technique	**Improvement**	**Reference**
Model Pruning	85ms inference time	[25]
Dynamic Protocols	**92%** attack mitigation	[26]
Federated Learning	**83%** data reduction	[24]

This synthesis of current research demonstrates that while AI has significantly advanced KD authentication (see Table V), realizing its full potential requires addressing persistent challenges through continued innovation and responsible implementation. The next decade promises further breakthroughs as the field matures toward enterprise-ready solutions.

D. Emerging Trends and Challenges

As keystroke dynamics continues to evolve, several emerging trends and challenges warrant attention. Privacy-preserving techniques address concerns related to sensitive biometric data by incorporating encryption and differential privacy, as proposed by Gupta et al. and Lee et al. [72], [73]. Multi-factor authentication systems, combining keystroke dynamics with other biometrics like facial recognition [74], offer enhanced security for critical applications. Security and privacy issues remain critical, particularly in mitigating risks from data breaches, as surveyed by Bhattasali et al. [75]. Mobile authentication presents unique challenges due to device variability and user mobility, necessitating adaptive models, as explored by Tsimperidis et al. [76]. These advancements highlight the need for interdisciplinary approaches to address scalability, privacy, and robustness in future keystroke dynamics systems.

V. CONCLUSION

After diving into 100 studies from 2015 to March 2025, it's clear that artificial intelligence has reshaped keystroke dynamics (KD) authentication in ways we couldn't have imagined a decade ago. Methods like deep learning and hybrid models—think LSTM and CNN-SVM—have slashed error rates by **40–60%** compared to the old statistical approaches. We're talking about systems hitting **98.7%** accuracy in free-text scenarios, which is a game-changer for real-world applications in banking, healthcare, and education. KD's strength lies in its ability to authenticate users continuously without needing fancy hardware, making it a practical, non-invasive option for securing sensitive systems. But it's not all smooth sailing. Our review uncovered some serious hurdles. Privacy is a big one—typing patterns can reveal a user's identity with an **87%** success rate, which is a real ethical minefield. Then there's the issue of security: GAN-based attacks can fool these systems **40%** of the time, exposing vulnerabilities that need urgent attention. On top of that, hardware limitations are a drag, with mobile devices often taking over 500ms to process, and touchscreens throwing in extra variability—flight times are 20% longer compared to physical keyboards. Looking forward, there's a lot of potential to iron out these kinks. Decentralized approaches like federated learning could cut down data exposure while keeping accuracy high. Dynamic defenses, such as challenge-response protocols, are already blocking **92%** of attacks in some tests. And if we can optimize for edge computing, we might halve latency, making KD viable on low-end devices. The path ahead will need collaboration across fields—computer science, ethics, and hardware engineering—to build systems that are not only secure and efficient but also respect user privacy. Standardizing protocols to meet industry benchmarks like FIDO will also be key to getting KD into mainstream use. This field is on the cusp of something big, but it'll take focused effort to make it a reality.

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